

Yiran Yang

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EDUCATION

University of Science and Technology of China (USTC)

Hefei, China

B.S. in Planetary Science, expected 2027 | GPA: 3.94/4.30 | Class rank: 1/79

GRANTS

Undergraduate Basic Research Project

2025-2026

National Natural Science Foundation of China (NSFC) | CNY 100,000

Selected as one of only 150 undergraduate recipients nationwide across all disciplines.

PUBLICATIONS

Zhang, C.[†], Yang, Y.[†], Cai, Y.[†], Zhang, S., He, C., Glein, C., Huang, F., Sherwood Lollar, B., & Hao, J. (2026). Dynamic controls on subsurface water chemistry and habitability on icy moons. *The Innovation*, 7(6), 101420. doi:10.1016/j.xinn.2026.101420

[†] Co-first authors; equal contribution.

RESEARCH EXPERIENCE

CH₄-H₂ Demixing in Ice-Giant Interiors with a Machine-Learned Potential

July 2026-Present

Advisor: Prof. Yunguo Li, USTC

- Developing a C-H Machine Learning Potential over 0.1-15 GPa and 400-2500 K on a SCAN+rVV10 DFT dataset.
- Using large-scale MLMD to construct the CH₄-H₂ phase diagram over 0.1-15 GPa and 400-2500 K.
- Coupling the phase relations and EOS to 1D Uranus and Neptune interior profiles to locate demixing layers and evaluate their effects on density structure, gravitational moments, and double-diffusive heat transport.

Space-Environment Controls on the Lunar Exosphere

Mar-Jul 2026

Advisors: Prof. Tong Dang and Prof. Jinting Kang, USTC

- Coupled lunar global MHD simulations across eight lunar phases with returned-sample statistical composition and sputtering/impact-vaporization models to quantify environmental controls on exosphere composition.
- Built a collisionless ballistic-transport and number-density framework using Liouville's theorem and velocity-space integration, revealing O-Si dominance and lower sputtering densities than predicted by uniform-flux models.

Fatty-Acid Vesicles as Protocell Compartments

Jul-Aug 2025

Summer Research Intern | Advisor: Prof. Sean Jordan, Dublin City University, Ireland

- Tested how silica affects fatty-acid vesicle formation and stability under simulated prebiotic Earth conditions.
- Quantified vesicle morphology and size distributions using UV-Vis spectroscopy, fluorescence and confocal microscopy, and dynamic light scattering.

Membraneless Polyester Microdroplets as Prebiotic Compartments

Dec 2024-Jun 2026

Advisors: Prof. Jihua Hao, USTC; Prof. Tony Jia, Earth-Life Science Institute (ELSI)

- Investigated how wet-dry cycles control the formation and properties of membraneless polyester microdroplets as potential prebiotic compartments.
- Characterized polyester chain-length distributions by mass spectrometry and evaluated microdroplet compatibility with biomolecules using fluorescence microscopy.

HONORS & AWARDS

National Scholarship, Ministry of Education of China (top 1%)	2023-2024
Yanbao Scholarship (top 1%)	2024-2025
Fundamental Discipline Top-Notch Talent Plan Scholarship	2023-2024
Fundamental Discipline Top-Notch Talent Plan Scholarship	2024-2025

SELECTED COURSEWORK

Physical Chemistry; Geochemistry; Astrobiology; Planetary Geology; Planetary Interior Physics; Planetary Atmospheres; Biochemistry; Microbiology

TECHNICAL SKILLS

Computational: Python, C; VASP, LAMMPS, DeePMD-kit, DP-GEN

Laboratory: ICP-MS, ICP-OES, XRD, FTIR, MALDI-TOF MS, UV-Vis spectroscopy, fluorescence microscopy, confocal laser scanning microscopy, dynamic light scattering

Languages: Mandarin Chinese (native); English (fluent; TOEFL iBT: 5.5/6.0)

SERVICE & OUTREACH

Conference Volunteer, Planetary Science Forum for Young Scientists	Mar 2025
Science Outreach Volunteer, Guizhou Province, China	Jul 2024
Conference Volunteer, International Workshop on Planetary Science	Jun 2024